

Stroke Prognosis & Clinical Scores

Clinical prediction scales for ischemic and hemorrhagic stroke outcomes.

STROKE PROGNOSIS SCALES

Ischemic Stroke: ASTRAL & PLAN

Hemorrhagic Stroke: ICH Score

ASTRAL Score (Acute Ischemic Stroke)

Predicts 90-day poor functional outcome (mRS > 2)

Predictor Variable	Points
Age	1 pt per 5 years
Severity (NIHSS)	1 pt per NIHSS pt
Time to admission >3h	2 pts
Range of visual fields (defect)	2 pts
Acute glucose (<3.7 or >7.3 mmol/L)	1 pt
Level of consciousness (impaired)	3 pts

Probability estimates are withheld pending validation. ASTRAL should not guide counseling of critically ill patients or treatment limitations.

PLAN Score (Acute Ischemic Stroke)

Original outcomes: 30-day mortality and death/severe dependency at discharge

Predictor Domain & Variables	Points
Preadmission dependence / Cancer	1.5 pts each
Preadmission CHF / Atrial Fibrillation	1.0 pt each
Level of consciousness (reduced)	5.0 pts
Age (decades)	1 pt per decade (max 10)
Neurologic: Leg / Arm weakness	2 pts each
Neurologic: Aphasia or Neglect	1.0 pt

Outcome percentages are withheld pending validation against the original cohort.

ICH Score (Intracerebral Hemorrhage)

Predicts 30-day mortality in spontaneous ICH

Predictor Component	Points
GCS Score: 3–4 (2 pts) 5–12 (1 pt) 13–15 (0 pts)	0–2 pts
Age: ≥ 80 years	1 pt
ICH Volume: ≥ 30 mL	1 pt
Intraventricular Hemorrhage (IVH): Present	1 pt
Infratentorial Origin of Hemorrhage	1 pt

Score vs. 30-Day Mortality Risk:

0	1	2	3	4	5
0%	13%	26%	72%	97%	100%

Historical cohort rates, not individual predictions. No percentage is assigned here to score 6; do not extrapolate a certain fatal outcome.

Modified Rankin Scale (mRS)

A widely used scale of global disability

Grade Clinical Description of Recovery State

- 0 No symptoms at all.
- 1 No significant disability despite symptoms; able to carry out all usual duties.
- 2 Slight disability; unable to carry out all previous activities but **independent**.
- 3 Moderate disability; requires some help but **able to walk unassisted**.
- 4 Moderately severe; **unable to walk or attend to bodily needs** without assistance.
- 5 Severe disability; bedridden, incontinent, requiring constant nursing care.
- 6 Dead.

Prognostication Principles & Limitations

- **Not for Care Limitations:** Historical scores describe severity and cohort outcomes, not individual recovery ceilings. In critically ill adults with ischemic stroke, admission NIHSS or ASTRAL alone is not reliable for poor-outcome counseling. The ICH Score can provide a general framework for communication, but no score should be the sole basis for an individual prognosis or for withholding reperfusion, decompression or life-sustaining treatment.
- **Serial Reassessment:** Integrate repeated examinations, imaging, treatment response and reversible confounders such as sedation. Observation and counseling timing are individualized; the first 24–72 hours are not a universal deadline for a reliable prognosis.
- **Acute ICH — BP and Hemostatic Treatment:** The ICH Score is not a treatment target. In mild-to-moderate spontaneous ICH presenting with SBP 150–220 mmHg, targeting 140 and maintaining 130–150 may be reasonable; lowering below 130 can be harmful. Intensive lowering has uncertain safety and benefit in large/severe ICH or surgical candidates, who need individualized management (AHA/ASA 2022). Recombinant factor VIIa (rFVIIa) given within 2h slowed hematoma growth but did **not** improve 180-day function and increased thromboembolic events (FASTEST, 2026; PMID 41653933) — **not** recommended for routine use.

Neuroprognostication: 2026 NCS/DGNI guidance, [PMID: 41942818](https://pubmed.ncbi.nlm.nih.gov/41942818/) (<https://pubmed.ncbi.nlm.nih.gov/41942818/>)

ASTRAL Score: Ntaios G, et al. *Neurology*. 2012;78(24):1916–22. [PMID: 22649218](https://pubmed.ncbi.nlm.nih.gov/22649218/) (<https://pubmed.ncbi.nlm.nih.gov/22649218/>)

PLAN Score: O'Donnell MJ, et al. *Arch Intern Med*. 2012;172(20):1548–56. [PMID: 23147454](https://pubmed.ncbi.nlm.nih.gov/23147454/) (<https://pubmed.ncbi.nlm.nih.gov/23147454/>)

ICH Score: Hemphill JC 3rd, et al. *Stroke*. 2001;32:891–7. [PMID: 11283388](https://pubmed.ncbi.nlm.nih.gov/11283388/) (<https://pubmed.ncbi.nlm.nih.gov/11283388/>)

ICH Management: AHA/ASA 2022 guideline, sections 5.1 and 7.1. [PMID: 35579034](https://pubmed.ncbi.nlm.nih.gov/35579034/) (<https://pubmed.ncbi.nlm.nih.gov/35579034/>)

mRS Scale: van Swieten JC, et al. *Stroke*. 1988;19:604–7. [PMID: 3363593](https://pubmed.ncbi.nlm.nih.gov/3363593/) (<https://pubmed.ncbi.nlm.nih.gov/3363593/>)